



ITA05-ACTIVITY

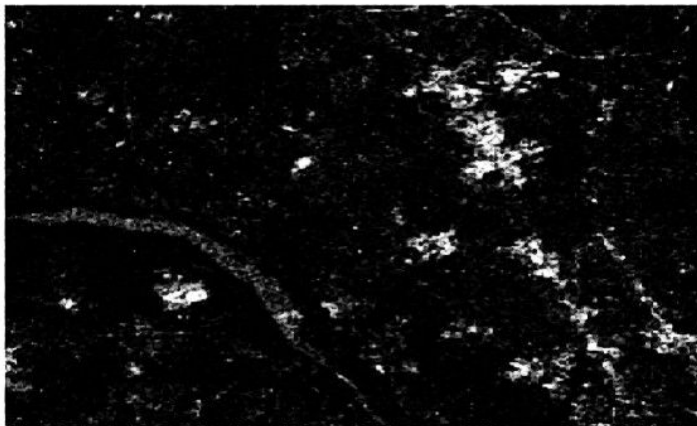
SCENARIO BASED ACTIVITY

Mention the techniques used with reason:

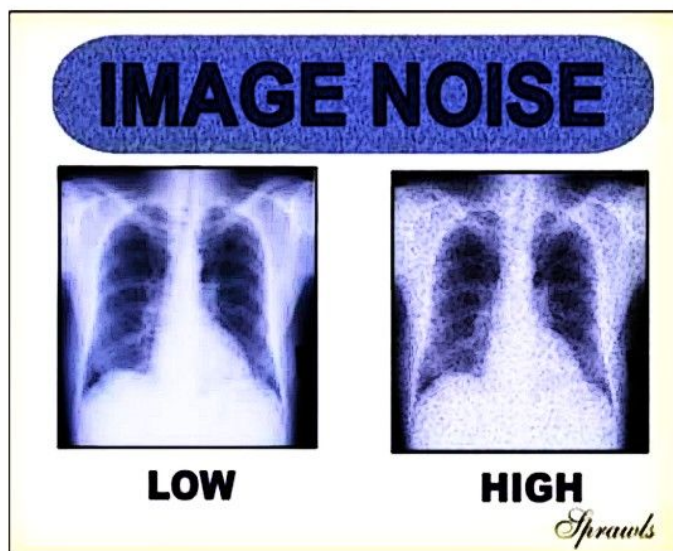
1. A CCTV image contains **salt-and-pepper noise** due to transmission errors.



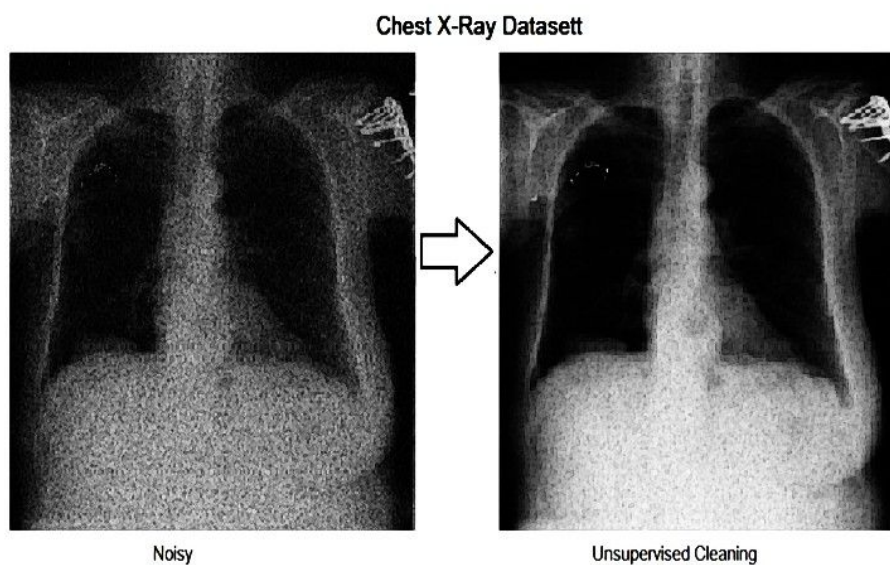
2. A satellite image is corrupted by **Gaussian noise** from sensors.



3. A medical X-ray image has random noise but fine edges must be preserved.



4. A scanned document has **black and white dots** scattered randomly.



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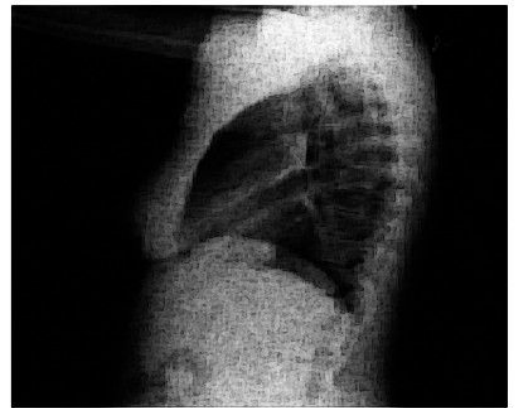
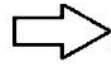


5. A low-light image shows heavy grain without clear boundaries.

Indiana University X-Ray Dataset

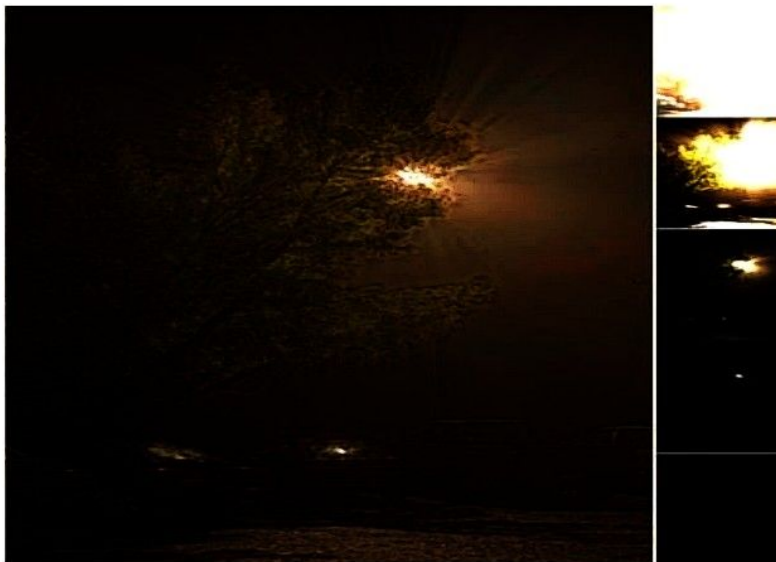


Noisy



Unsupervised Cleaning

6. A foggy road image has **very low contrast**.



7. An underwater image appears dull and washed out.

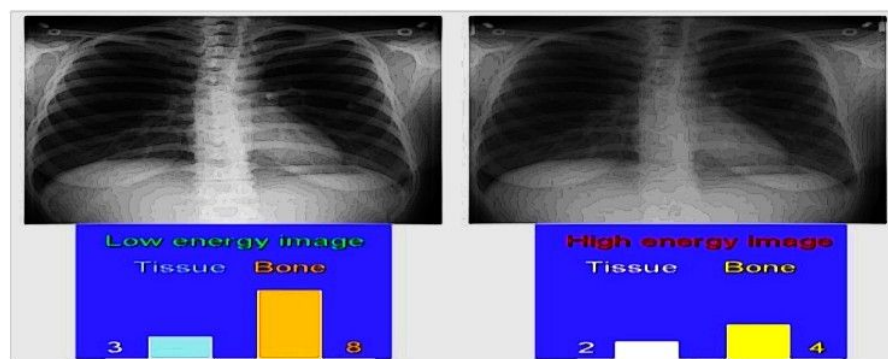


8. A chest X-ray shows poor visibility of internal structures.

7. An underwater image appears dull and washed out.



8. A chest X-ray shows poor visibility of internal structures.



9. A night-time traffic image lacks intensity variation.



10. A grayscale image uses only a narrow intensity range.



Activity-2 Co

- 1) Median filter: It is used to remove Salt and Pepper noise from CCTV Image.
It preserves image details while eliminating noisy pixels.
- 2) Gaussian filter: It is used to reduce Gaussian noise in Satellite Images. It smooths the image & minimizes sensor noise.
- 3) Bilateral filter: It removes random noise in satellite images, it smooths while preserving fine edges. It is ideal for medical x-ray images.
- 4) Median filter: It removes randomly scattered black and white dots in scanned documents. It improves visibility.
- 5) Gaussian filter: It reduces heavy graining in low light images. It smooths the image and improves visibility.
- 6) Histogram Equalization: It enhances the contrast of foggy road images and makes them more visible.
- 7) It Improves Contrast Stretching: The brightness of the contrast of under water images. It makes the image appear clearer.
- 8) Clutter: It enhances local contrast in chest x-ray images. It improves visibility of internal structure without noise.

9) Histogram Equalization

* It increases intensity variation in Night-time
faded Image. This improves overall image visibility

10) Contrast Stretching:-

* It expands the narrow intensity range of Gray
Scale Image. This enhances the overall image
contrast

